SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ANSWER SCRIPT EVALUATION SHEET

CONTINUOUS INTERNAL ASSESSMENT 1

Programme	B.E / B.Tech	Course Code	CSA07
Course Name:	Computer Networks	Date	3-11-2026
Duration:	1: 30 Hr	Marks	50

Q.No.	Max. Marks	Awarded marks
Q1	10	10
Q2	10	05
Q3	10	09
Q4	10	05
Q5	10	10
Total	50	39

Total in Words:

Thirty Nine

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Programme: B.E / B.Tech

Course Code: CSA07

Course Name: Computer Networks

Duration: 1: 30 Hr

Date : 03.11.2026 **Max Marks: 50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

EVALUATION KEY

Q1. OSI Model – Data Link and Transport Layers (CO1, BL2 & BL4) – 10 Marks

(a) Data Link and Transport Layer Functions and Reliability – 6 Marks

Expected point	Evaluation guideline	Marks
Data Link layer	Correct description of Data Link layer functions.	2
Transport layer	Correct description of Transport layer functions.	2
Collective reliability	Clear explanation of how the two layers together ensure end-to-end and hop-by-hop reliability.	2

Model answer

- **Data Link Layer (Layer 2):** Handles node-to-node delivery, framing, physical (MAC) addressing, and error detection over a single physical link.
- **Transport Layer (Layer 4):** Manages end-to-end communication between application processes, segmentation, and end-to-end reliability (acknowledgments, sequence numbers and retransmissions).
- **Collective reliability:** The Data Link layer ensures error-free transmission across each individual hop, while the Transport layer ensures reliable, complete deliveryfrom the original source host to the final destination host across the entire path.

(b) Connection-Oriented vs. Connectionless Services – 4 Marks

Expected point	Evaluation guideline	Marks
Differentiation	Clear differentiation between connection-oriented and connectionless services.	2
Examples	Accurate real-world communication examples for both.	2

Model answer

- **Connection-Oriented:** Establishes a formal logical connection (handshake) before data transmission. Guarantees delivery and order. Example: **TCP**; placing a telephone call.
- **Connectionless:** Transmits data immediately without prior connection setup. No delivery guarantees at this layer. Example: **UDP**; sending a postal letter.

Q2. Encapsulation and Network Layer Routing (CO1, BL2) – 10 Marks (a) Encapsulation and Decapsulation – 6 Marks

Expected point	Evaluation guideline	Marks
Encapsulation	Detailed step-by-step description of encapsulation at the source.	3
Decapsulation	Detailed step-by-step description of decapsulation at the destination.	3

Model answer

- **Encapsulation:** As data moves down the OSI layers at the source, each layer adds its own control header (and trailer at the Data Link layer).

Layer	Unit of data	Added at this layer
Application / upper layers	Data	Application data
Transport	Segment	Transport header
Network	Packet	Network header (IP addresses)
Data Link	Frame	Header and trailer (MAC addresses)
Layer	**Unit of data**	**Added at this layer**
Physical	Bits	Signals on the medium

- **Decapsulation:** As bits are received and moved up the OSI layers at the destination, each layer strips off its corresponding header/trailer, processes it, and passes the payload up to the next layer. **(b) Network Layer Routing and Logical Addressing – 4 Marks**

Expected point	Evaluation guideline	Marks
Layer identified	Correct identification of the Network layer.	1
Routing	Clear explanation of packet routing across interconnected networks.	1.5
Logical addressing	Accurate explanation of how logical addressing functions.	1.5

Model answer

- **Layer responsible:** Network Layer (Layer 3 – IPv4/IPv6).
- **Logical addressing:** Assigns unique hierarchical logical IP addresses containing network and host portions. Routers inspect the destination IP address in the packet header to make forwarding decisions across interconnected networks.

Q3. IP Addressing and Subnetting (CO1,CO2, BL2) – 10 Marks

(a) Classful Addressing Limitations and CIDR – 6 Marks

Expected point	Evaluation guideline	Marks
Classful limitations	Clear explanation of classful IP limitations (address wastage, scalability issues).	3
CIDR	Clear explanation of how CIDR/VLSM overcomes these limitations.	3

Model answer

- **Classful limitations:** Rigid fixed classes (A, B, C) led to severe address wastage and very large global routing tables because of inflexible block sizes.
- **CIDR:** Removes class boundaries by using flexible prefix lengths (variable masks). This allows address blocks of theright size andsupports route aggregation to keeproutingtables small. **(b) Subnet Calculation – 4 Marks**

Expected point	Evaluation guideline	Marks
Network ID	Correct Network ID calculation.	1.5
Broadcast address	Correct Broadcast Address calculation.	1.5
Usable range	Correct usable host IP range.	1

Model answer

- Given: 192.168.25.130, mask 255.255.255.192 (/26). Block size in the last octet = 256 − 192 = 64. The host address 130 lies in the block 128–191.
- **(i) Network ID:** 192.168.25.128 • **(ii) Broadcast Address:** 192.168.25.191
- **(iii) Usable host range:** 192.168.25.129 to 192.168.25.190 **Q4.**

Subnetting 172.16.0.0/16 (CO2, BL3) – 10 Marks Given network 172.16.0.0/16; at least 6 subnets required.

(a) Subnet Mask – 3 Marks

Expected point	Evaluation guideline	Marks
Bits / CIDR	Correct bit calculation and CIDR notation (/19).	1.5
Dotted-decimal mask	Correct dotted-decimal subnet mask (255.255.224.0).	1.5

Model answer

- Bits borrowed: $2^3 = 8 \geq 6$, so **3 bits** are borrowed. New prefix: **/19**.
- **Dotted-decimal subnet mask:** 255.255.224.0

(b) Total Subnets and Usable Hosts – 3 Marks

Expected point	Evaluation guideline	Marks
Total subnets	Correct calculation of total subnets.	1.5
Usable hosts	Correct calculation of usable host IPs per subnet.	1.5

Model answer

- **Total subnets:** $2^3 = 8$.
- **Usable hosts per subnet:** host bits = 32 − 19 = 13, so $2^{13} - 2 =$ **8,190**. **(c) First Two Subnets – 4 Marks**

Expected point	Evaluation guideline	Marks
Subnet 1	Accurate Network ID, usable range and broadcast for Subnet 1.	2
Subnet 2	Accurate Network ID, usable range and broadcast for Subnet 2.	2

Model answer

Block size in the 3rd octet = 256 − 224 = 32.

Subnet	Network ID	First usable IP	Last usable IP	Broadcast
1	172.16.0.0	172.16.0.1	172.16.31.254	172.16.31.255
2	172.16.32.0	172.16.32.1	172.16.63.254	172.16.63.255

Q5. Routing in Network Simulation Tools (CO2, BL3 & BL4) – 10 Marks

(a) Static vs. Dynamic Routing and Preferences – 6 Marks

Expected point	Evaluation guideline	Marks
Comparison	Clear comparison/contrast of static and dynamic routing.	3
Conditions	Accurate identification of operational conditions favouring static routing.	3

Model answer

- **Static routing:** Manually set by administrators. Secure and with zero routing-protocol overhead, but does not adapt automatically to failures.
- **Dynamic routing:** Automatically discovers paths using protocols (OSPF, EIGRP). Adapts to topology changes, but consumes CPU and bandwidth.
- **When static is preferred:** Small, stub networks with a single exit point (e.g., an ISP connection), where simplicity, control and zero overhead are required.

(b) Router CLI Configuration Steps – 4 Marks

Expected point	Evaluation guideline	Marks
Command syntax	Correct mode entry and command syntax for static routing (ip route …).	2.5
Execution flow	Proper execution flow (global configuration mode, saving/exiting).	1.5

Model answer

```
Router> enable
Router# configure terminal
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)# end
Router# write memory
```

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CIA - 1

Course name : Computer Networks
Course code : CSA0710

① a) Explain the Data link and Transport layers.

Datalink layer:

The data link layer provides communication b/w directly connection devices its main function are.

framing: Divides data into frames
MAC Addressing: identifies devices on a local network.
Error Detection: Detect errors using CRC.
Flow Control: Controls the speed of data transmission
Access Control: Manages access to the communication medium.

Transport layer:

The Transport layer provides communication b/w application on different devices.

Segmentation: Divides data into smaller segments.
Port Addressing: Identifies applications.
Error Recovery: Prevents receiver overload.
Reliable Delivery: TCP Provides overloaded and reliable data delivery.

b)

Connection-oriented	Connectionless
> Establishes a connection first > TCP provides reliable delivery Ex: file transfer using TCP	> Does not require connection setup > UDP does not guarantee delivery Ex: DNS queries using UDP.

a)

a) Encapsulation and Decapsulation:

Encapsulation: Each OSI layer adds it header as data moves from sender to receiver's network

Decapsulation: the receiver removes headers layer by layer to obtain the original data.

Order: Data -> Segment -> Packet -> Frame -> Bits.

b) Network Layer:

The network layer performs routing b/w networks. It uses logical IP addresses to identify destination and forwards packet through routers.

a) Classful Addressing and CIDR

classful addressing uses fixed network size it wastes IP addresses and provides limit flexibility CIDR user flexible Prefix lengths such as 126. it reduces address wastage and supports route aggregation.

b) subnet calculation:

Given IP: 192.168.25.130
mask : 255.255.255.192 (126)
network ID: 192.168.25.128.
Broadcast : 192.168.25.191
first usable : 192.168.25.129.
Last usable : 192.168.25.190
usable hosts : 62

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(4) a) subnet mask:

Given: 172.16.0.0/16 & at least 6 subnets

$2^3 = 8$ subnets

Brrowed bits = 3
new CIDR : /19
Subnet mask : 255.255.224.0

b) Subnets and Hosts

Total subnets : $2^3 = 8$

Host bits : $32 - 19 = 13$

Usable hosts : $2^{13} - 2 = 8190$

c) First two subnets:

Details	Subnet 1	Subnet 2
Network ID	172.16.0.0	172.16.32.0
First Usable	172.16.0.1	172.16.32.1
Last Usable	172.16.31.254	172.16.63.254
Broad Cast	172.16.31.255	172.16.63.255

(5) a) Static vs Dynamic Routing:

Static routing : Routes are configured manually suitable for small, stable networks.

Dynamic routing: Routes are learned automatically using protocols such as RIP and OSPF.

-) Static routing uses fewer resources while dynamic routing adapts to network changes.

b) static route Configuration:

Router > enable
Router # Configure terminal
Router (Config) # ip route 192.168.2.0
~~Router C~~
255.255.255.0 . 10.0.0.2
Router (config)# end
Router # show ip route :.